

The Blade and the Halting Principle: A Mathematical Proof of the Existence of “Atoms” in an Ideal Cutting Model

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Abstract

We introduce a simple formal model of an *ideal imaginary blade* that acts on portions of matter as a partition operator. We define a well-founded measure quantifying the divisibility of each portion and assume a natural axiom: **every non-trivial cut strictly decreases this measure**. Under these assumptions we prove that every portion admits a finite sequence of cuts that ends in an indivisible element—an **atom**.

We provide a modal-logic formalization, show the relation with the halting problem, and include a prototype Isabelle/HOL formalization sketch.

1. Introduction

The idea of “atom” (from Greek *a-tomos*, “uncuttable”) can be modeled with mathematical precision. Imagine a perfect blade that can cut any piece of material. A natural question arises:

■ *Can such a blade cut forever, or must the process eventually stop?*

We show that if we accept an extremely mild assumption—that each cut reduces a well-founded measure—then infinite cutting is impossible. A terminal point must exist: an **atom**.

This provides a simple mathematical justification of atomism without relying on physics or experimental evidence.

2. Model and main theorem

2.1 Set of portions

Let:

- $\mathcal{P}$: the set of abstract portions of matter.

2.2 The blade

A cut is an operation:

$$Cut(x, y_1, \dots, y_k)$$

meaning that cutting x yields sub-portions $y_1, \dots, y_k$ with $k \geq 2$.

2.3 Divisibility measure

We define:

$$\mu : \mathcal{P} \rightarrow S$$

where S is a **well-founded** set under the relation $\prec$.

Example: $S = \mathbb{N}$ with $<$.

2.4 Axiom of descent

For every non-trivial cut:

$$Cut(x, y_1, \dots, y_k) \Rightarrow \mu(y_i) \prec \mu(x) \text{ for all } i.$$

This formalizes the intuitive idea that *each cut reduces something*.

2.5 Definition (atom)

A portion a is an **atom** if it admits no non-trivial cut:

$$Atom(a) \equiv \neg \exists y_1, \dots, y_k. k \geq 2 \wedge Cut(a, y_1, \dots, y_k).$$

3. Main theorem

Theorem (Cut–Halting).

$\forall x \in \mathcal{P} \exists a \in \mathcal{P} \exists n \in \mathbb{N} : \text{after } n \text{ cuts starting from } x, a \text{ is an atom.}$

That is: **every cutting process ends in an atom.**

4. Proof

Fix $x \in \mathcal{P}$. Consider:

$$M = \{\mu(z) \mid z \text{ is reachable from } x \text{ by finitely many cuts}\}.$$

Since S is well-founded, M contains minimal elements.

Let a be a portion with minimal measure in M .

Assume a can be cut:

$$\text{Cut}(a, y_1, \dots, y_k).$$

By descent:

$$\mu(y_i) \prec \mu(a) \quad \forall i,$$

contradicting the minimality of $\mu(a)$.

Thus a is an **atom**.

□

5. Computational interpretation: halting

The blade behaves like a computation:

- each cut is a computational step,
- the measure μ is a ranking function,
- ranking functions that strictly decrease guarantee **termination**.

Thus:

■ *An atom is a halting state.*

Infinite cutting is impossible under the model's assumptions.

6. Modal-logic formalization

6.1 Worlds

- Worlds W : states of the cutting process.
- Accessibility relation R : one cutting step.

$$wRw' \equiv \text{"w' is obtained by cutting in w"}.$$

6.2 Modal operator

$$\Diamond_{\text{cut}} P_x \equiv \text{"x exists in some cut-successor world"}.$$

6.3 Modal definition of atom

$$\text{Atom}_w(x) \equiv \neg \Diamond_{\text{cut}} P_x.$$

6.4 Modal theorem

In any Kripke frame with a well-founded ranking function that strictly decreases along R , all accessibility branches are finite $\rightarrow$ every object eventually becomes an atom.

7. Constructive variant with $\mathbb{N}$

If:

$$\mu : \mathcal{P} \rightarrow \mathbb{N}$$

and every cut satisfies:

$$\mu(y_i) < \mu(x),$$

then the length of the cutting process is bounded by $\mu(x)$.

By complete induction, an atom is reached in fewer than $\mu(x)$ steps.

8. Isabelle/HOL formalization sketch

Suggested file name:

`Cut_Halting_Model.thy`

```
theory Cut_Halting_Model
imports Main
begin

typedcl Obj
typedcl world

consts R :: "world  $\Rightarrow$  world  $\Rightarrow$  bool"
consts Sub :: "world  $\Rightarrow$  Obj  $\Rightarrow$  Obj  $\Rightarrow$  bool"
consts Cut :: "world  $\Rightarrow$  Obj  $\Rightarrow$  Obj set  $\Rightarrow$  bool"
consts mu :: "world  $\Rightarrow$  Obj  $\Rightarrow$  nat"

definition Atom :: "world  $\Rightarrow$  Obj  $\Rightarrow$  bool" where
  "Atom w x  $\equiv$   $\neg$  ( $\exists w'$ . R w w'  $\wedge$  ( $\exists y$ . Sub w' y x  $\wedge$  y  $\neq$  x))"

axiomatization where
  descent:
    "[ Cut w x Y; y  $\in$  Y; R w w' ]  $\Rightarrow$  mu w' y < mu w x"

inductive reachable :: "world  $\Rightarrow$  world  $\Rightarrow$  bool" where
  r_refl: "reachable w w"
| r_step: "[ R w u; reachable u v ]  $\Rightarrow$  reachable w v"

theorem cut_halting:
  fixes w x
  shows " $\exists w'$ . reachable w w'  $\wedge$  Atom w' x"
  sorry

end
```

9. Discussion

- The model captures the original Greek intuition of atomism.
- The existence of atoms follows not from physics but from **logic**.
- The argument is equivalent to the impossibility of infinite descending chains.
- Computationally:
 - “blade” = transition function,
 - “atom” = halting state,
 - “measure” = ranking,
 - “termination” = finiteness of cutting.

This gives an elegant, purely mathematical justification for atomism.

10. Conclusion

In any system where:

1. every cut decreases a well-founded measure,
2. cuts correspond to computational steps,
3. no infinite descending chains exist,

it follows that **every object reaches an indivisible state**.

Atomism emerges as a logical theorem:

a halting theorem disguised as geometry of cuts.